

Animesh Saha

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EDUCATION

University at Buffalo, The State University of New York <i>Bachelor of Science, Computer Science — Specialization in Artificial Intelligence</i> – GPA: 3.85, Summa Cum Laude Tau Beta Pi Inductee (Spring 2025) Dean's List (Spring 2025, Spring 2024, Fall 2023)	Buffalo, New York <i>Expected December 2025</i>
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SKILLS

Languages: Python, Java, OCaml, SQL, JavaScript, TypeScript, HTML/CSS
Frameworks/Tools: React.js, Next.js, FastAPI, LangChain, PyTorch, Docker, Qdrant, FAISS, sentence-transformers, Google Gemini API, Firebase, MySQL, Pandas, NumPy
Concepts: RAG & Vector Search, LLM Evaluation, REST API Design, Data Structures & Algorithms, Computer Vision, Deep Learning, OAuth 2.0/Authentication, Agile Development

EXPERIENCE

Software Engineer (Capstone) <i>Capstone Project</i> – Engineered mood-based music recommendation system with a full-stack REST API and React frontend, improving recommendation accuracy by 60%. – Integrated OAuth 2.0 authentication and MySQL database integration, automating Spotify playlist generation for 2,500+ users.	August 2024 – December 2024
Undergraduate Teaching Assistant — CSE 305: Programming Languages <i>Dept of CSE, University at Buffalo</i> – Debugged complex programming projects and taught functional programming and language-theory concepts to 250 students, grading assignments and resolving technical questions in person and online.	August 2025 – December 2025
Academic Tutor <i>Tutoring and Academic Support Services, University at Buffalo</i> – Tutored 60+ students one-on-one and staffed drop-in tutoring for 600+ students in Data Structures, Algorithms, and Calculus 2, visualizing algorithms (BFS, DFS, Dijkstra, A*) to build deep understanding.	August 2024 – December 2025
Student Assistant <i>Special Programs Management, School of Engineering and Applied Science, University at Buffalo</i> – Led a team of 5 to coordinate Dean's Advisory Council meetings and reviewed 50+ resumes while conducting interviews to select qualified candidates for open positions.	August 2023 – December 2025

PROJECTS

RAG System over arXiv Research Papers <i>Python, Docker, Qdrant, LangChain, FastAPI</i> – Built an end-to-end Retrieval-Augmented Generation (RAG) ingestion pipeline over 200+ arXiv research papers, chunking abstracts and full-text PDFs into 2,344 retrievable chunks, embedded with sentence-transformers and indexed in a Dockerized Qdrant vector database. – Engineered a hand-wired retrieval-to-generation loop integrating the Gemini API, achieving a 77.8% retrieval hit-rate@5 on an 18-question hand-labeled evaluation set, and built a dual LLM-as-judge/human evaluation harness that uncovered a self-grading ceiling-effect bias in the LLM judge. – Refactored the pipeline with LangChain (LCEL) while holding retrieval logic identical via a custom retriever wrapper, producing a quantified framework-vs-hand-wired tradeoff analysis, and diagnosed a payload-schema bug in LangChain's default vector-store integration before it caused silent citation data loss. – Built a FastAPI backend exposing the pipeline through a Pydantic-validated REST endpoint and a Next.js/React/TypeScript chat frontend rendering grounded answers with clickable citations to source papers.	July 2026
Ice Breaker — Real-Time Social Platform <i>Next.js, React, Firebase</i> – Built a real-time, anonymous group-chat web app used live by 1,000+ attendees at a large in-person social gathering, requiring no login or personal data storage. – Integrated Gemini LLM to summarize conversations and answer contextual questions within group chats. – Optimized Next.js/React architecture, reducing latency from 500ms to under 200ms in high-traffic conditions.	November 2025
Draw It Right — Winner, UB Hacking 2024 <i>Python, PyTorch, OpenCV, Mediapipe</i> – Designed a finger-tracking, gesture-based virtual drawing tool using OpenCV/Mediapipe, and integrated a PyTorch neural network to classify user-drawn objects in real time with 92.5% accuracy. – Deployed the app on Streamlit for an interactive, user-friendly interface, winning UB Hacking 2024.	November 2024